

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5	1
1 1 2 3 4	

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2 int main() {
3     int n;
4     scanf("%d", &n);
5     int arr[1000];
6     int i, j;
7     for (i = 0; i < n; i++) {
8         scanf("%d", &arr[i]);
9     }
10    for (i = 0; i < n; i++) {
11        for (j = i + 1; j < n; j++) {
12            if (arr[i] == arr[j]) {
13                printf("%d", arr[i]);
14                return 0;
15            }
16        }
17    }
18    return 0;
19 }
20
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

The first line contains T, the number of test cases. Following T lines contain:

- Line 1 contains N1, followed by N1 integers of the first array
- Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6	10 57
2 7 10 15 57 246	

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2 #include <stdlib.h>
3 void findIntersection() {
4     int t;
5     if (scanf("%d", &t) != 1) return;
6     while (t--) {
7         int n1;
8         scanf("%d", &n1);
9         int *arr1 = (int *)malloc(n1 * sizeof(int));
10        for (int i = 0; i < n1; i++) {
11            scanf("%d", &arr1[i]);
12        }
13        int n2;
14        scanf("%d", &n2);
15        int *arr2 = (int *)malloc(n2 * sizeof(int));
16        for (int i = 0; i < n2; i++) {
17            scanf("%d", &arr2[i]);
18        }
19        int i = 0, j = 0;
20        int first = 1;
21        while (i < n1 && j < n2) {
22            if (arr1[i] == arr2[j]) {
23                if (!first) printf(" ");
24                printf("%d", arr1[i]);
25                first = 0;
26                i++;
27                j++;
28            } else if (arr1[i] < arr2[j]) {
29                i++;
30            } else {
31                j++;
32            }
33        }
34        printf("\n");
35        free(arr1);
36        free(arr2);
37    }
38 }
39
40 int main() {
41     findIntersection();
42     return 0;
43 }
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

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Find the intersection of two sorted arrays.

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Input:

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Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1	10 57
3 10 17 57	
6	
2 7 10 15 57 246	

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2 #include <stdlib.h>
3 void solve() {
4     int t;
5     if (scanf("%d", &t) != 1) return;
6     while (t--) {
7         int n1;
8         if (scanf("%d", &n1) != 1) break;
9         int *arr1 = (int *)malloc(n1 * sizeof(int));
10        for (int i = 0; i < n1; i++) {
11            scanf("%d", &arr1[i]);
12        }
13        int n2;
14        if (scanf("%d", &n2) != 1) {
15            free(arr1);
16            break;
17        }
18        int *arr2 = (int *)malloc(n2 * sizeof(int));
19        for (int i = 0; i < n2; i++) {
20            scanf("%d", &arr2[i]);
21        }
22        int i = 0, j = 0;
23        int first = 1;
24        while (i < n1 && j < n2) {
25            if (arr1[i] == arr2[j]) {
26                if (!first) {
27                    putchar(' ');
28                }
29                printf("%d", arr1[i]);
30                first = 0;
31                i++;
32                j++;
33            } else if (arr1[i] < arr2[j]) {
34                i++;
35            } else {
36                j++;
37            }
38        }
39        putchar('\n');
40        free(arr1);
41        free(arr2);
42    }
43 }
44 int main() {
45     solve();
46     return 0;
47 }

```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2 int main()
3 {
4     int n, k;
5     int a[1000];
6     scanf("%d", &n);
7     for (int i = 0; i < n; i++)
8         scanf("%d", &a[i]);
9     scanf("%d", &k);
10    int i = 0;
11    int j = 1;
12    while (i < n && j < n)
13    {
14        if (i == j)
15        {
16            j++;
17        }
18        else if (a[j] - a[i] == k)
19        {
20            printf("1");
21            return 0;
22        }
23        else if (a[j] - a[i] < k)
24        {
25            j++;
26        }
27        else
28        {
29            i++;
30        }
31    }
32    printf("0");
33    return 0;
34 }
35
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

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Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k, i \neq j$.

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1 - If pair exists

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14        if (i == j)
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